MILITARY SPECIFICATION

DETONATOR, FLASH, M35
LOADING, ASSEMBLING, AND PACKING

1. SCOPE

1.1 This specification covers the quality assurance provisions and requirements not covered by the drawings for the loading, assembling and packing for one type of flash detonator designated as Detonator, Flash, M35.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL–P–116 —Preservation, Methods of.
MIL–A–2550 —Ammunition and Special Weapons, General Specification for.

STANDARDS

MILITARY

MIL–STD–105D —Sampling Procedures and Tables for Inspection by Attributes.
MIL–STD–109 —Quality Assurance Terms and Definitions.
MIL–STD–1235 —Single and Multi-level Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ORDNANCE CORPS

7548099 —Box, Packing, Ammunition, for Detonators.
7548103 —Box, Set-Up, Packing, Ammunition, Outer, for Boosters, Detonators or Primers. Assembly, Details, Packing and Marking.
7548104 —Box, Set-Up, Packing, Ammunition, Inner, for Boosters, Detonators and Primers. Assembly, Details, Packing and Marking.
8797780 —Detonator, M35, Loading Assembly.

PUBLICATIONS

ORDNANCE CORPS

IEL–8797780 —Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with applicable drawings and specifications.

3.2 Detonators. The detonators shall comply with all requirements specified on Drawing (dwg.) 8797780, and with all re-

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quirements specified in applicable specifications.

3.3 **Moisture content of lead azide.** The moisture content of the lead azide used in loading the detonators shall not exceed 0.50 percent when tested as specified in 4.3.1.

3.4 **Sensitivity.** The detonators shall function on the first attempt and shall cause perforation of the lead disc to a diameter of not less than ¼ inch when tested as specified in 4.3.3.

3.5 **Workmanship.** All parts shall be free of chips, dirt, grease, rust and other foreign material. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 **General quality assurance provisions.** The supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL–STD–109 in order to define terms used herein. The provisions of Specification MIL–A–2550 shall apply.

4.1.1 *Contractor quality assurance system.* The contractor shall provide and maintain an effective quality assurance system in compliance with the requirements of Specification MIL–I–45208. The contractor's quality assurance system shall contain a procedure for the reporting of critical defectives found during the course of his inspection and during inspection by the Government. Any critical defect encountered, unless annotated by an asterisk (*) to the right of the characteristic code number (see 6.2), will be reported within 48 hours. A monthly summary report covering those critical defects annotated by an asterisk will be furnished. The agencies to whom the reports will be furnished are identified in paragraph 6. The reports shall contain, as a minimum, the following information:

(a) Contractor and contract number.
(b) Number and revision date of specification.
(c) Characteristic code number.
(d) Lot or serial number.
(e) Number of defectives found.
(f) Assignable cause or analysis of occurrence.
(g) Corrective action taken or recommended.

When the corrective action is beyond the control of the contractor or the scope of the contract, the report shall contain the contractor's recommendation to the Government.

4.1.2 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.
(b) Number of units of product inspected.
(c) Results obtained for all inspection performed.
(d) Drawing, specification number and date, together with an identification and date of changes.
(e) Certificates of analyses on all material procured directly by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.

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(f) Number of items in the lot.

(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 *Government verification.* Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be in accordance with Specification MIL–I–45208 and will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions.

4.2.1 *Lot formation.* A lot shall consist of detonators produced by one manufacturer in one unchanged process in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, or fit, as determined by the Government, shall not necessitate changing the lot interfix number. Each lot shall contain:

(a) Parts bearing the same lot interfix number from one manufacturer.

(b) Lead azide from not more than one lot.

(c) Tetryl from not more than one lot.

4.2.2 *Examination.* Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL–STD–105D except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Standard MIL–STD–1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Assembly, prior to loading (see dwg. 8797780).

Categories	*Defects*	*Method of inspection*	*Code No. (see 6.2).*
Critical:	None defined.		
Major:	AQL 0.25 percent		
101.	Coating compound or lacquer on exposed surface of disc	Visual	01001
Minor:	AQL 0.40 percent		
201.	Evidence of poor workmanship (see 3.5)	Visual	01002

4.2.2.2 Assembly (see dwg. 8797780).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:			
1.	Cup split or cracked	Visual	02001*
Major:	AQL 0.65 percent		
101.	Length	Gage	02002
102.	Diameter	Gage	02003
103.	Color of lacquer incorrect (either end)	Visual	02004
104.	Not crimped full 360 degrees	Visual	02005
105.	Disc loose	Manual	02006

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106.	Disc punctured or torn	Visual	02007
107.	Cup damaged	Visual	02008
Minor:	AQL 0.40 percent		
201.	Evidence of poor workmanship (see 3.5)	Visual	02009

4.2.2.3 Box, set-up (see dwg. 7548104).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Box damaged exposing contents	Visual	03001
102.	Contents loose	Manual	03002
Minor:	AQL 0.40 percent		
201.	Marking missing or incorrect	Visual	03003

4.2.2.4 Box, set-up in barrier bag (see dwg. 7548103).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.25 percent		
101.	Barrier bag damaged	Visual	04001
102.	Seal not permitting minimum subsequent seals	Visual	04002
Minor:	AQL 0.40 percent		
201.	Marking missing or incorrect	Visual	04003

4.2.2.5 Box, packing, sealed (see dwg. 7548099).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Board broken	Visual	05001
102.	Strapping missing or loose	Visual	05002
103.	Marking missing, misleading or unidentifiable	Visual	05003
Minor:	None defined.		

4.2.3 *Testing.*

4.2.3.1 *Moisture content of lead azide.* (See 3.3)—Major defect—Code No. 02010. The moisture content of the lead azide shall be determined for each lot at the time of loading. A sample of 2 grams shall be selected and tested as specified in 4.3.1. If the moisture content is in excess of the percentage specified and if loading has not begun, the lot of lead azide shall be rejected until proper corrective action has been accomplished as verified by repeating the moisture content test specified in 4.3.1. If the loading has begun, action on the unloaded lead azide shall be as specified above and all detonators loaded with the questionable lead azide shall be rejected.

4.2.3.2 *Binder-lubricant content of tetryl.* (See dwg. 8797780.)—Major defect—Code No. 02011. The binder-lubricant content of the tetryl shall be determined for each batch or lot at the time of loading. An accurately weighed sample of 5 grams shall be selected and tested as specified in 4.3.2. If the binder-lubricant content is in excess of the percentage specified, and if loading has not begun, the lot of tetryl shall be rejected until proper corrective action has been accomplished as verified by repeating the binder-lubricant content test specified in 4.3.2. If the loading has begun, action on the unloaded tetryl shall be as specified above and all detonators loaded with the questionable tetryl shall be rejected.

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4.2.3.3 *Sensitivity.* (See 3.4)—Major defect—Code No. 02012. A sample of 300 units shall be selected from each lot for this test. Testing shall be as specified in 4.3.2 using the applicable test equipment of 4.2.4. Failure of one, or more, units of the sample in meeting either requirement shall reject the lot.

4.2.3.4 *Heat seal or quick leak test of barrier bag, as applicable.* (See dwg. 7548103.) —Major defect—Code No. 04004. The test procedures for the test shall be as prescribed in Specification MIL–P–116. Sampling and acceptance criteria shall be in accordance with inspection level S–4 of Standard MIL–STD–105D.

4.2.4 *Inspection equipment.* Index of Inspection Equipment Number IEL 8797780 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.2.4.1 *Government rights to documentation.* Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement of parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and wherever inspection equipment drawings are needed.

4.2.4.2 *Supply and maintenance.* Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL–I–45607.

4.2.4.3 *Government use of contractor's inspection and test equipment.* The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.3 Test methods and procedures.

4.3.1 *Determination of moisture content of lead azide.* A tared dried aluminum dish 2 inches in diameter shall be covered with a sheet of polytetrafluoroethylene 2½ inches square and sufficiently flat to form a tight cover. The accurately weighed sample shall be transferred to the moisture dish and the dish and contents shall be transferred to an oven and dried at 95 degrees Centigrade (C.) plus or minus 2 degrees C. for 2 hours. The cover shall be placed in the sulfuric acid desiccator during the drying period. After drying, the dish shall be quickly transferred to the desiccator and allowed to cool to room temperature. The polytetrafluoroethylene cover shall be placed on the dish and the assembly weighed. The loss in weight shall be calculated to percentage of moisture in the lead azide to determine compliance with 3.3 as follows:

$$\text{percentage of moisture} = \frac{(A-B)}{A} \times 100$$

where:

A = the weight of the sample before drying.

B = the weight of the sample after drying.

4.3.2 *Determination of binder-lubricant content of tetryl.* The accurately weighed sample shall be transferred to a tared medium porosity sintered glass filtering crucible. The sample shall be extracted with four 10 milliliter (ml.) portions of acetone, maintaining each portion in contact with the residue for approximately 5 minutes before applying suction. Any lumps which may persist shall be broken up and the sides of the crucible washed down with each 10 ml. portion of acetone to insure complete removal of the tetryl. The residue shall be aspirated until the odor of acetone is no longer detectable. The crucible and contents shall be placed in a drying oven at 93 to 97 degrees C. for ½ hour, cooled in a desiccator and weighed. The above extraction shall be repeated, dry-

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ing and weighing with individual 10 ml. portions of acetone until the loss in weight on subsequent extractions totals less than 0.005 gram. The weight of residue in the crucible shall be calculated to the percentage of total binder-lubricant.

4.3.3 *Sensitivity.* The detonator shall be assembled to the applicable test fixture specified in 4.2.4. A steel ball $1\frac{3}{16}$ inch in diameter and weighing 3.95 ounces, maximum, shall be dropped on the firing pin to function an M55 Pimer from a height of 8 inches, maximum. The height of drop shall be measured as the height from the top of the firing pin to the bottom of the ball. A detonator that fails to fire or fails to create a hole in the lead disc of the required size shall be classed defective.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Levels B (see 6.4) and C. The detonator shall be preserved, marked and packaged in accordance with dwgs. 7548103 and 7548104.

5.2 Packing and marking.

5.2.1 *Level C.* The detonators shall be packed and marked in accordance with dwg. 7548099.

5.3 Data cards. Data card information shall be prepared for each lot in accordance with Specification MIL–A–2550.

6. NOTES

6.1 Ordering data. Procurement documents shall specify the following:

(a) Title, number and date of this specification.

6.2 Inspection code numbers. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of 6.3.1 through 6.3.7.

6.3.1 *Inspection equipment lists.* Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (OO Form 1242–3) the numbers of drawings of existing equipment designs or codes as indicated in 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "contractor design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 *Inspection equipment list codes.* The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5, and 6.3.6 will be designated in the Inspection Equipment List by the following codes:

CDOF —Contractor's design responsibility on Ordnance format; ORDM–608–12.
CDCF —Contractor's design responsibility on contractor format.
OD —Ordnance design.
ODMU —Ordnance design, mandatory for use.
CE —Commercial equipment.
SCD —Specification Control Drawing.

6.3.3 *Ordnance designs.* Ordnance designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Ordnance design is listed for an item of inspection equipment. Ordnance designs fall into two basic classifications; mandatory and nonmandatory. When an Inspection Equipment List references mandatory Ordnance designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with nonmandatory designs and

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with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 *Specification control drawings.* Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagrammatic, or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 *Commercial equipment.* Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and government. Contractors are not required to furnish drawings of commercial inspection equipment but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

6.3.6 *Contractor designs.* Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDOF shall comply with the format and requirements of ORDM 608–12, and, in addition, contain the detail and information specified above.

6.3.7 *Submission of contractor designs.* Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–ND. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

6.4 Level B packaging. In the event the expected storage time of the detonators is in excess of 180 days, the barrier bag shall be used as specified on the applicable drawing. Expected storage time shall be determined by the procuring activity.

6.5 Check test for possible deterioration. The following test is designed to prevent the loading of deteriorated detonators into a major ammunition item. The test should be performed on detonators from each lot which has been stored under normal conditions for more than two years, or under adverse conditions for any period of time. If the specification for the ammunition item into which the detonators are loaded does not require that the check test be performed, the contracting officer should insert a provision for the performance of the check test in the contract. The contracting officer shall arrange to have the contractor reimbursed for any expenses incurred in the performance of the check test for possible deterioration.

6.5.1 *Test for sensitivity.* A sample of 300 detonators shall be selected from the shipment. In the event 2 or more lots are repre-

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sented in the shipment, a sample of 300 shall be taken from each lot and each sample shall be tested separately. Testing shall be as specified in 4.3.2. Failure of four or more units of a sample in meeting either requirement shall reject the detonators represented by that sample.

6.6 Submission of critical defect reports. For U. S. Army Munitions Command procurements, one copy of each critical defect report will be furnished the cognizant Government inspection agency; the Army Ammunition Procurement and Supply Agency, ATTN: SMUAP–RO and Picatinny Arsenal, ATTN: SMUPA–ND.

6.6.1 *Reporting to other services.* For other than U. S. Army Munitions Command procurements, critical defect reporting shall be as prescribed by the Contracting Officer.

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